

SC2001/CE2101/CZ2101:

Algorithm Design and Analysis

Appendix

(Merge operation in Mergesort)

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Courtesy of Dr. Ke Yiping, Kelly's slides

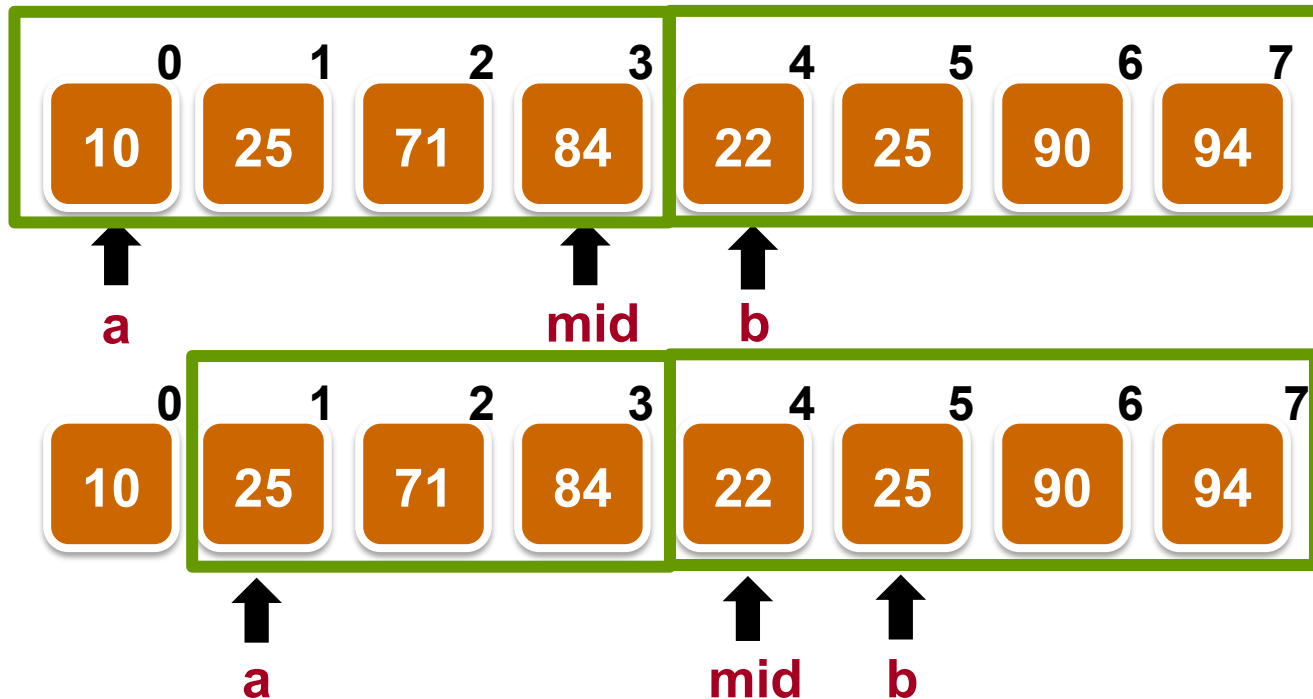
Merge Function

```
void merge(int n, int m)
{
    int mid = (n+m)/2;
    int a = n, b = mid+1, i, tmp;
    if (m-n <= 0) return;
    while (a <= mid && b <= m) {
        cmp = compare(slot[a], slot[b]);
        if (cmp > 0) { //slot[a] > slot[b]
            tmp = slot[b++];
            for (i = ++mid; i > a; i--)
                slot[i] = slot[i-1];
```

Merge Function

```
        slot[a++] = tmp;
    } else if (cmp < 0) //slot[a] < slot[b]
        a++;
    else { //slot[a] == slot[b]
        if (a == mid && b == m)
            break;
        tmp = slot[b++];
        a++;
        for (i = ++mid; i > a; i--)
            slot[i] = slot[i-1];
        slot[a++] = tmp;
    }
} // end of while loop;
} // end of merge
```

Merge Operation



Parameters for merge:

$n:0, m:7$

$\text{mid} = (0+7)/2 = 3;$

$a = n; b = \text{mid}+1;$

Comparison:

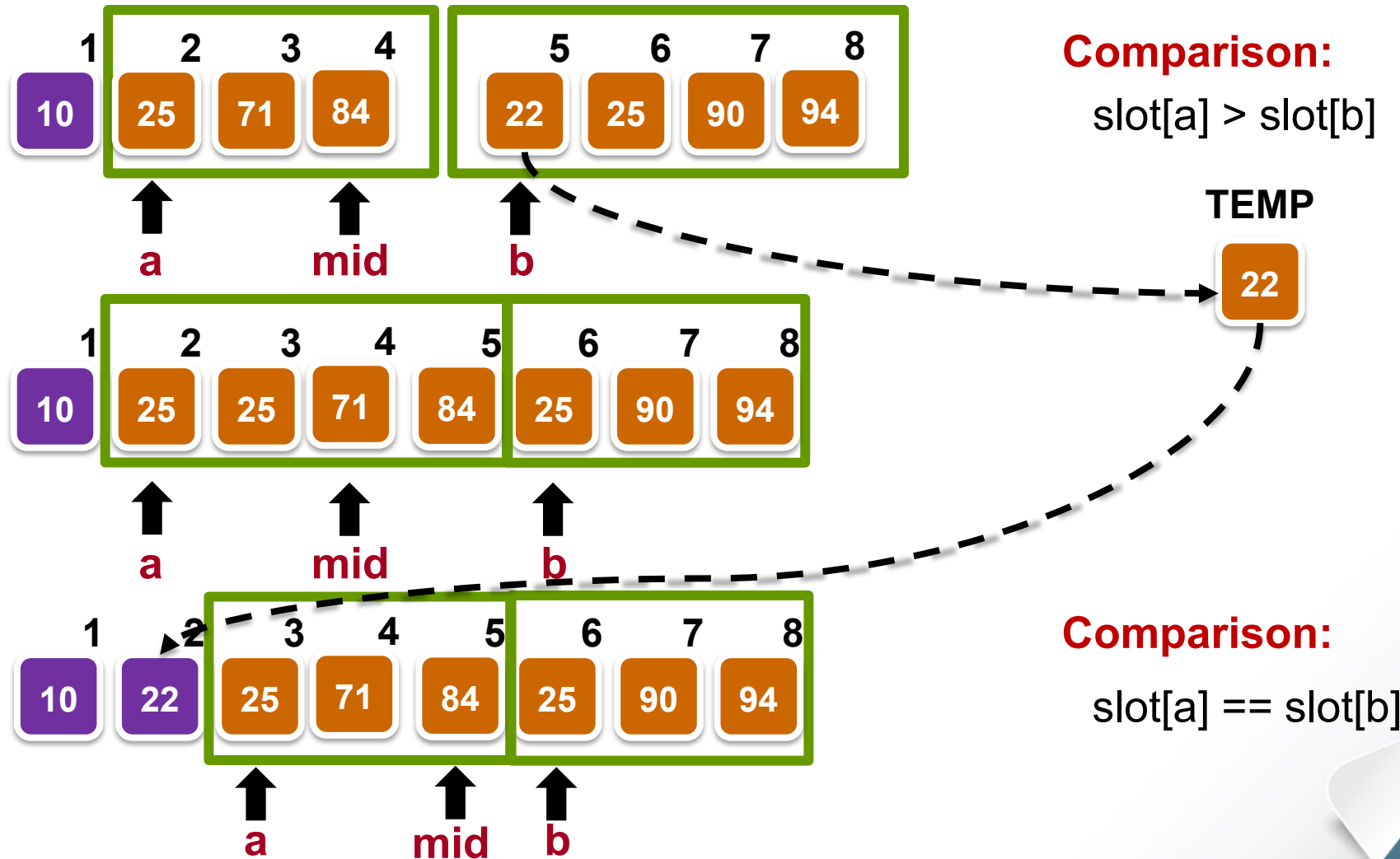
$\text{slot}[a] < \text{slot}[b]$

a : the 1st element of the 1st half

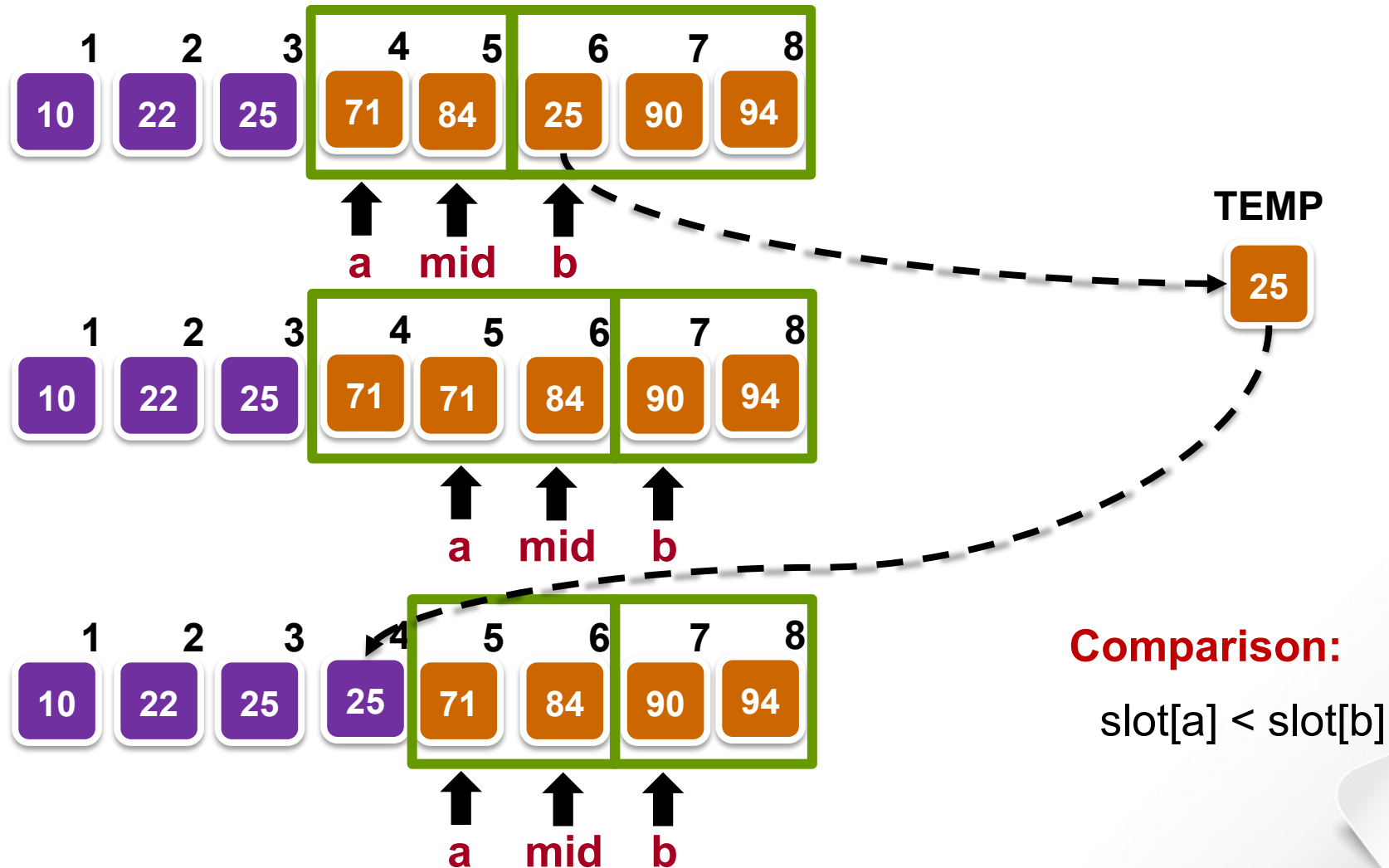
mid : the last element of the 1st half

b : the 1st element of the 2nd half

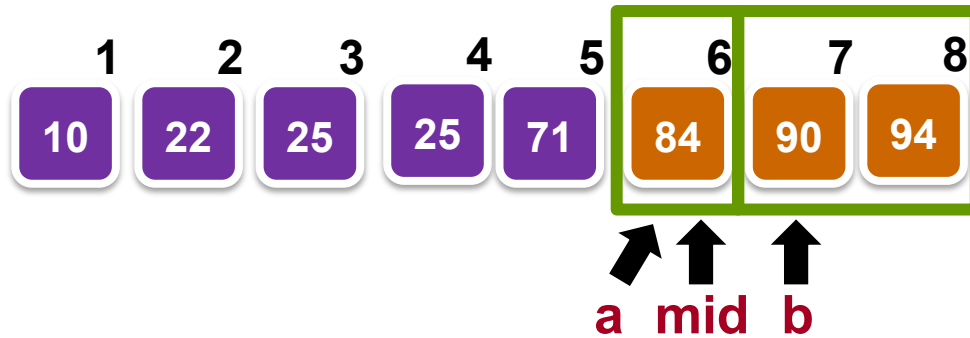
Merge Operation



Merge Operation

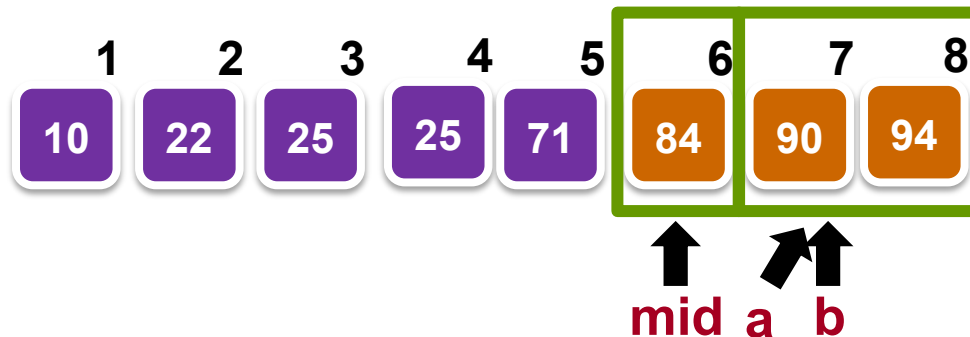


Merge Operation



Comparison:

$\text{slot}[a] < \text{slot}[b]$



1st half empty

Merge operation completed